
DinoV2

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Content

Self-supervised
Learning

DinoV2

실험 결과

1. Self-supervised Learning
2. DinoV2
3. 실험 결과

Introduction

Self-supervised Learning

- 단일 개념 Class로 예측하지 않아, 더 풍부한 학습이 가능함
- Self-supervised Learning ViT는 다양한 task에 활용 가능함

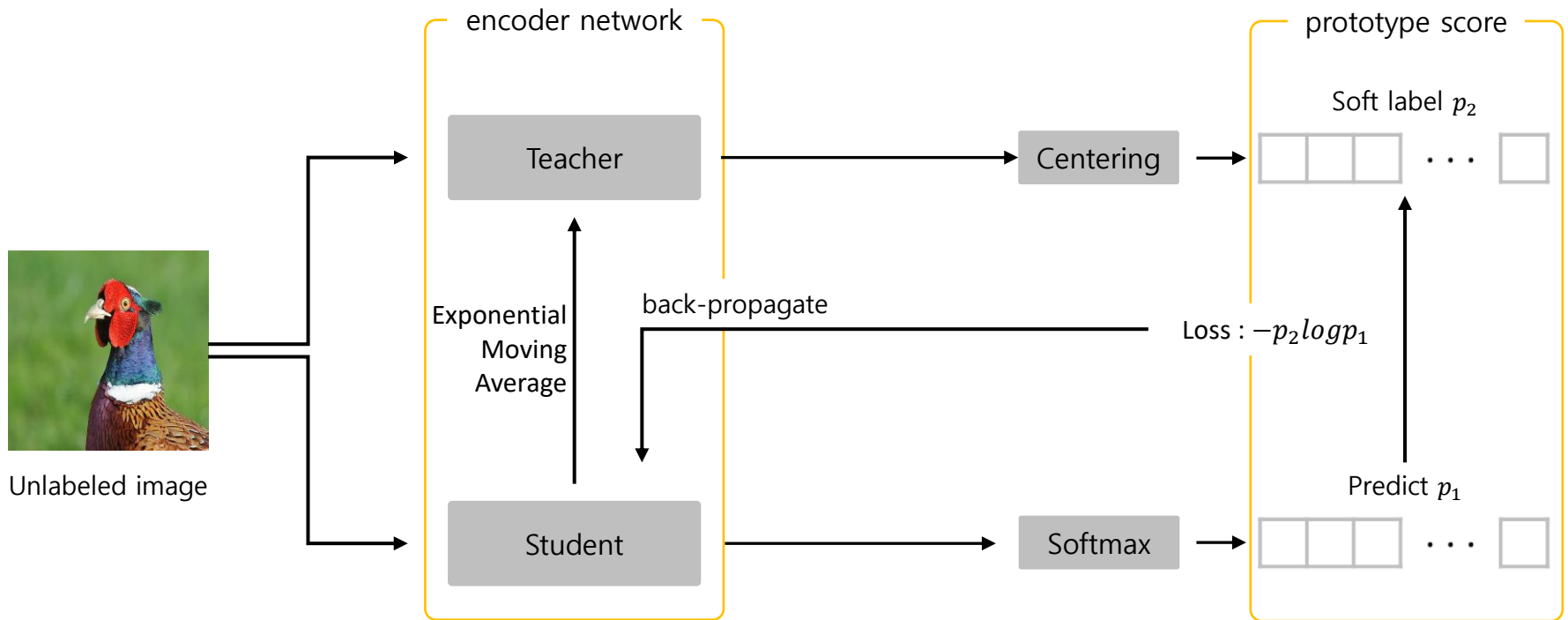
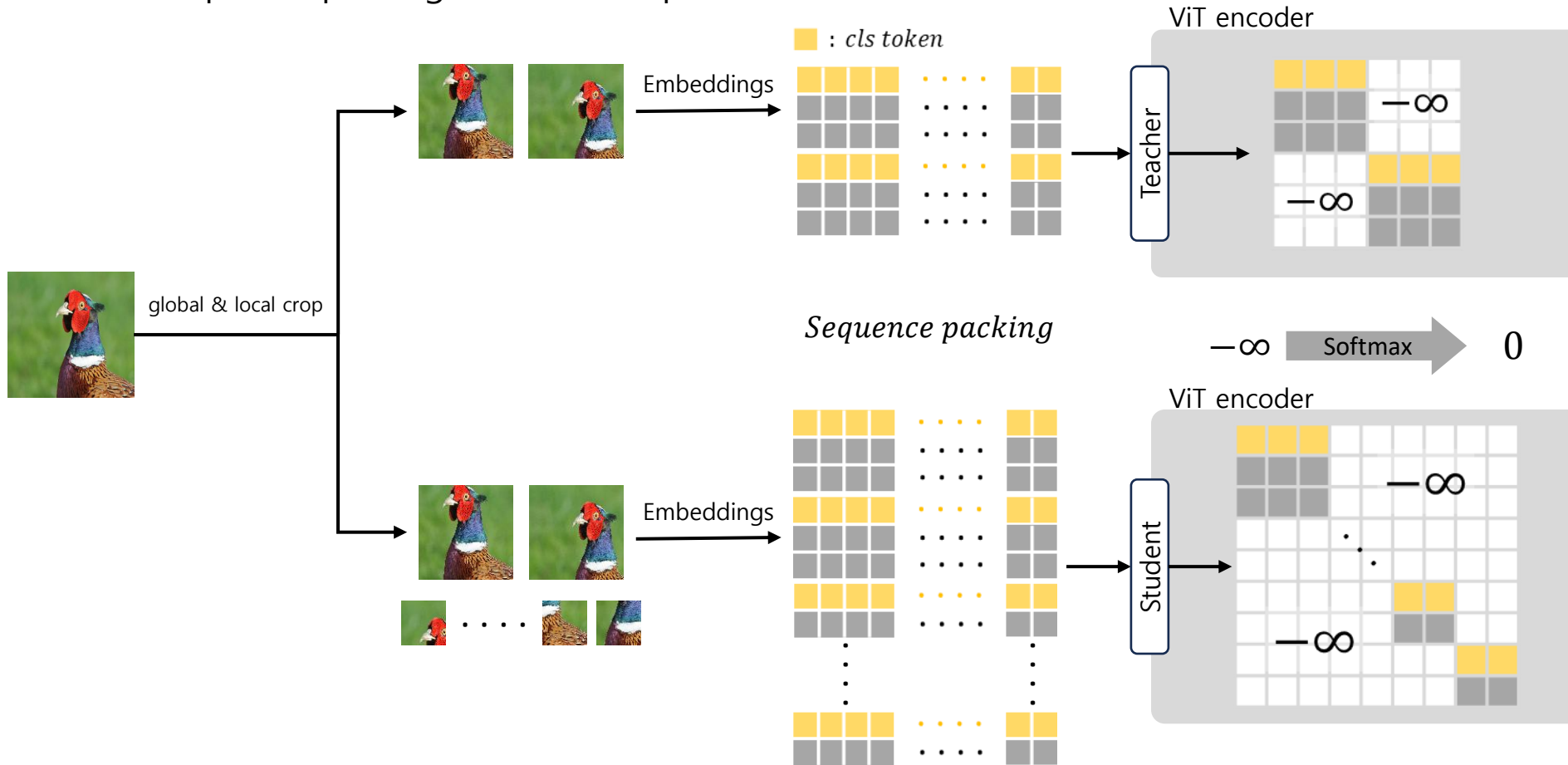


Image-level Objective

- Local image로 Global image의 맥락을 맞추도록 학습
- Sequence packing trick으로 Sequence를 concatenate하여 연산 효율성 증가

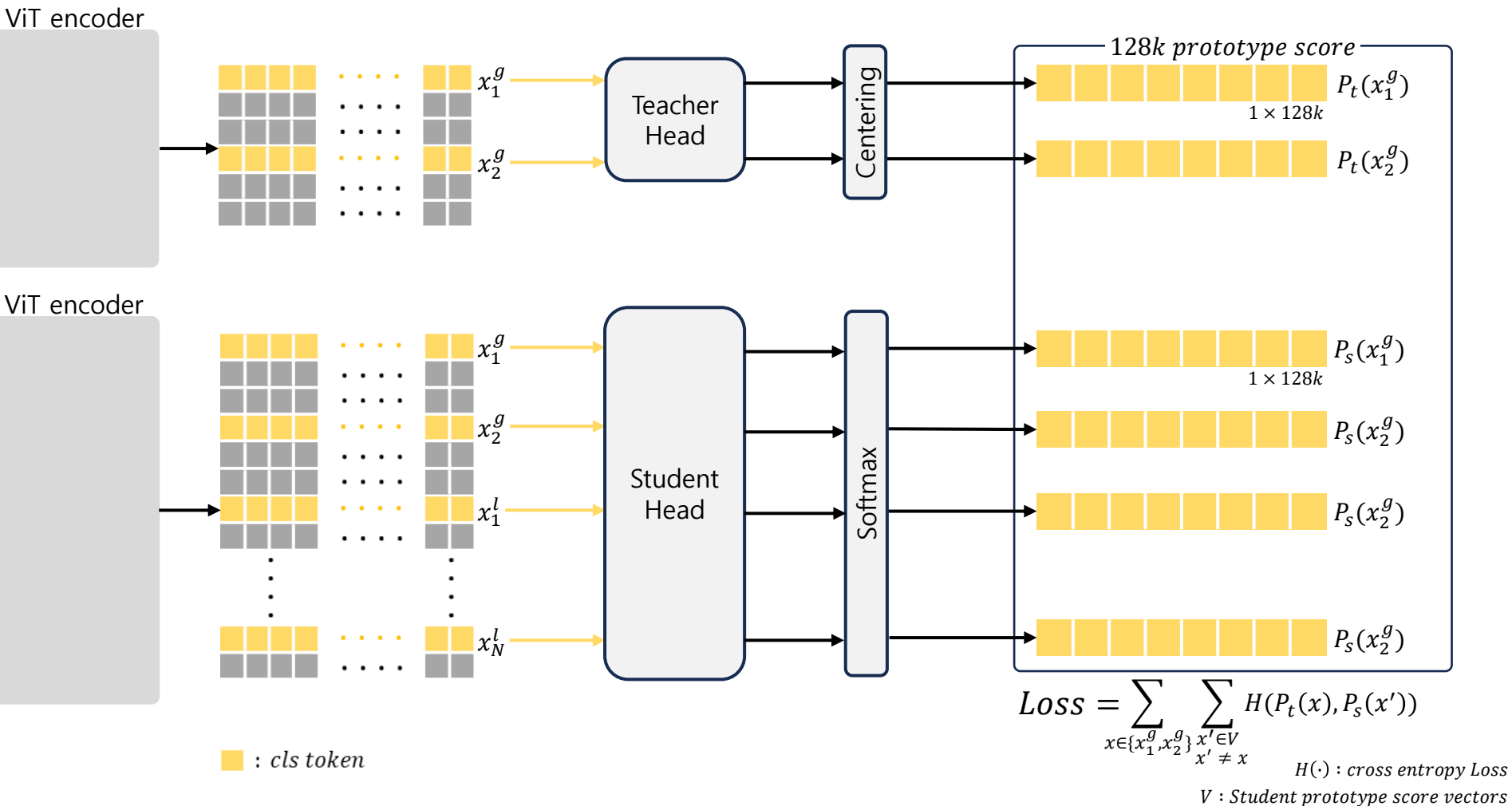


[1] M. Oquab, T. Darcet, et al. "Dinov2: Learning robust visual features without supervision." *In Computer Vision and Pattern Recognition*. 2023

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Image-level Objective

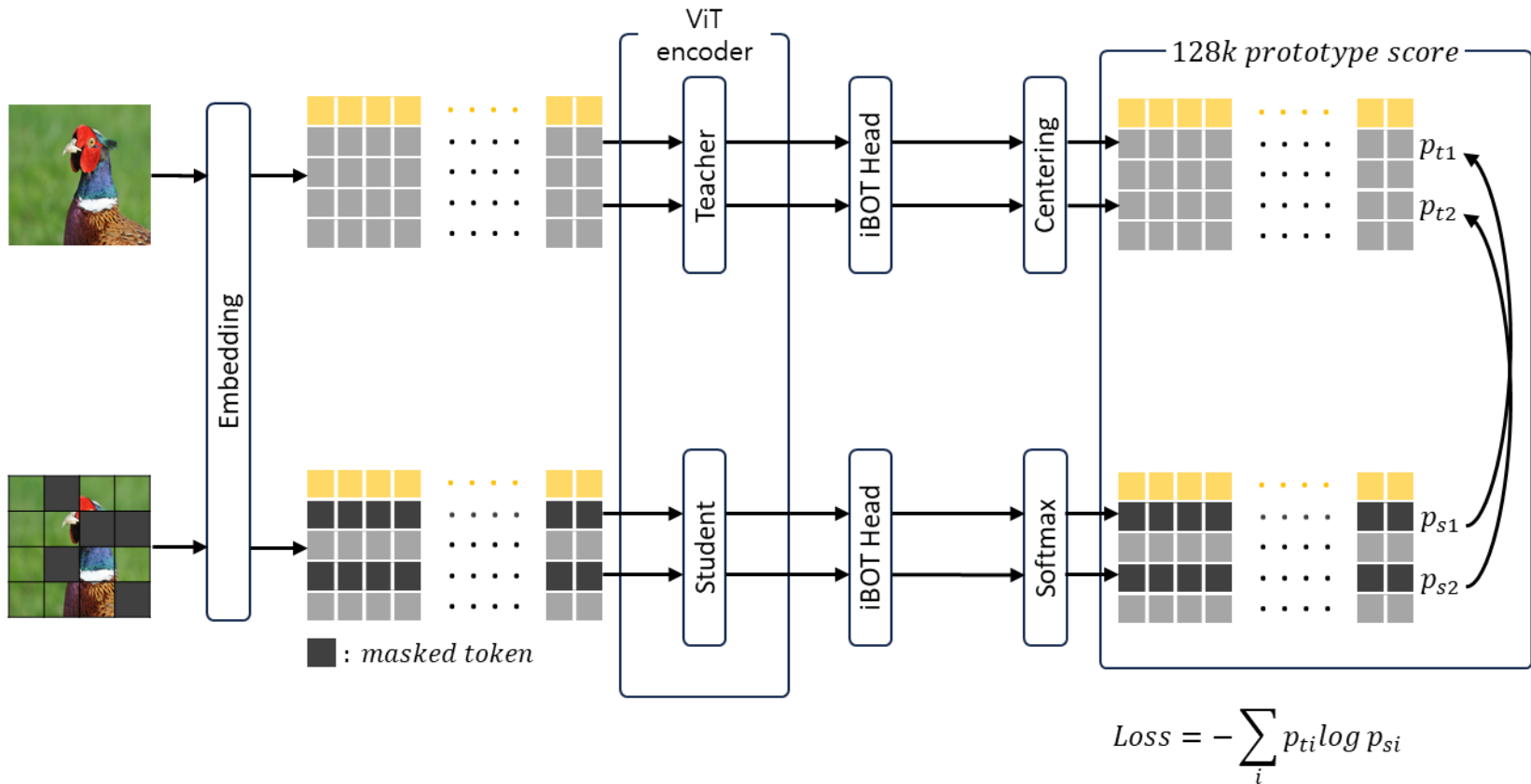
- Student score가 Teacher score와 일치하도록 학습



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▪ Patch-level objective

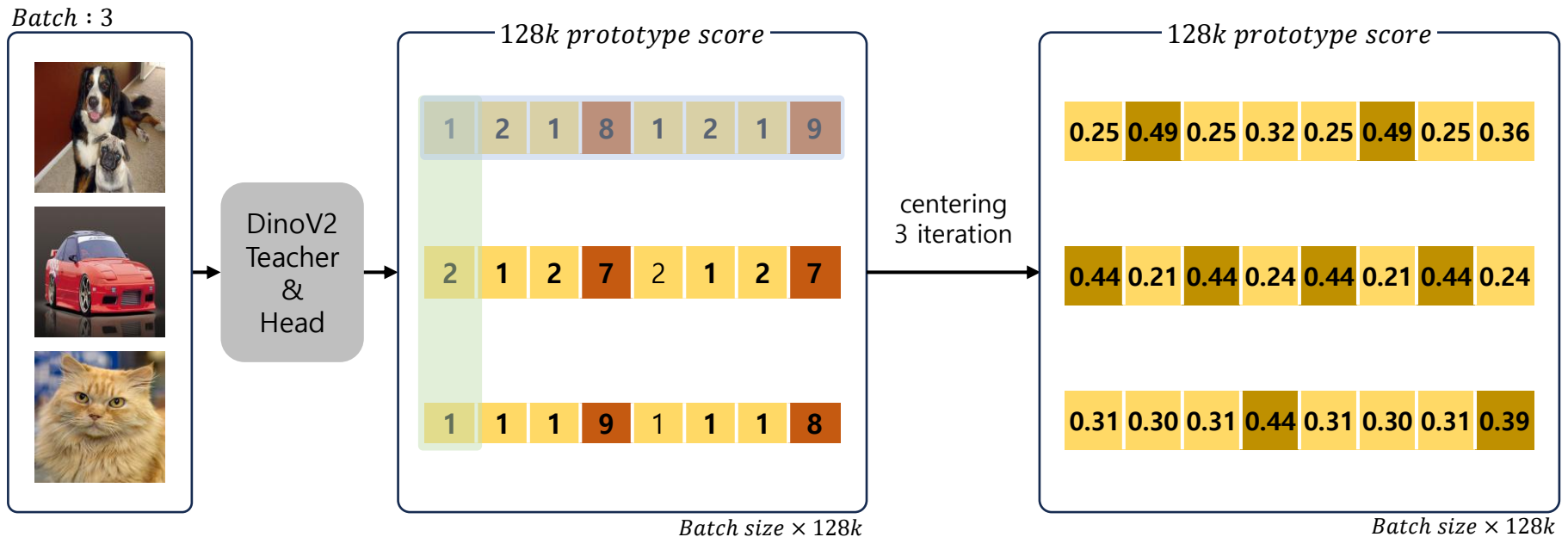
- Mask token의 정보를 예측하도록 학습
- 주변 token의 정보로 masked token의 semantics 정보 학습



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▪ Sinkhorn-Knopp centering

- Batch 내의 모든 이미지에 대해 같은 정답을 내지 않게 하기 위해 사용
- Prototype score를 균등하게 분할하기 위한 제약



Experiment

▪ Setting

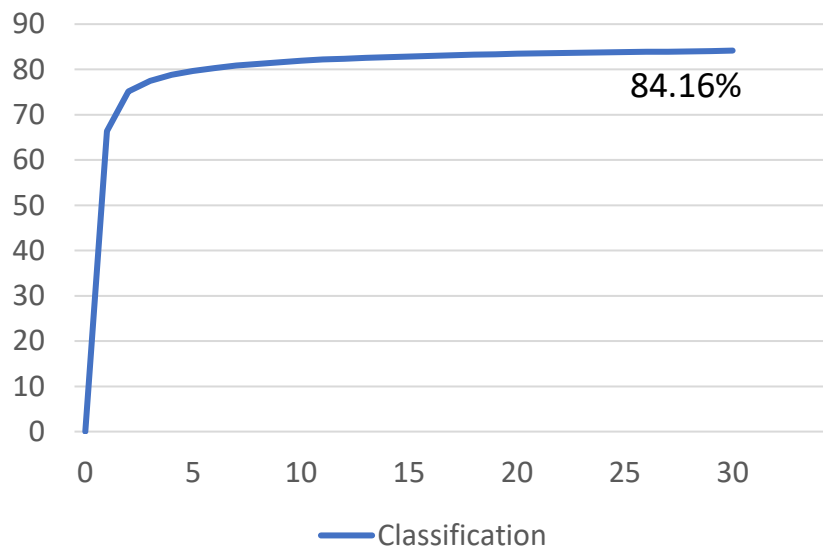
	Classification	Segmentation
Pre-trained model	DinoV2-ViT-Base/14	DinoV2-ViT-Base/14
Pre-trained dataset	LVD-142M	LVD-142M
Dataset	TinyImageNet	PASCAL VOC2012
Epoch	30	30
Batch size	4096	64
Learning rate	0.01	0.01
Optimizer	SGD	Adam
Augmentation	Random Horizontal Flip Random Crop	Random Horizontal Flip Random Crop

Result

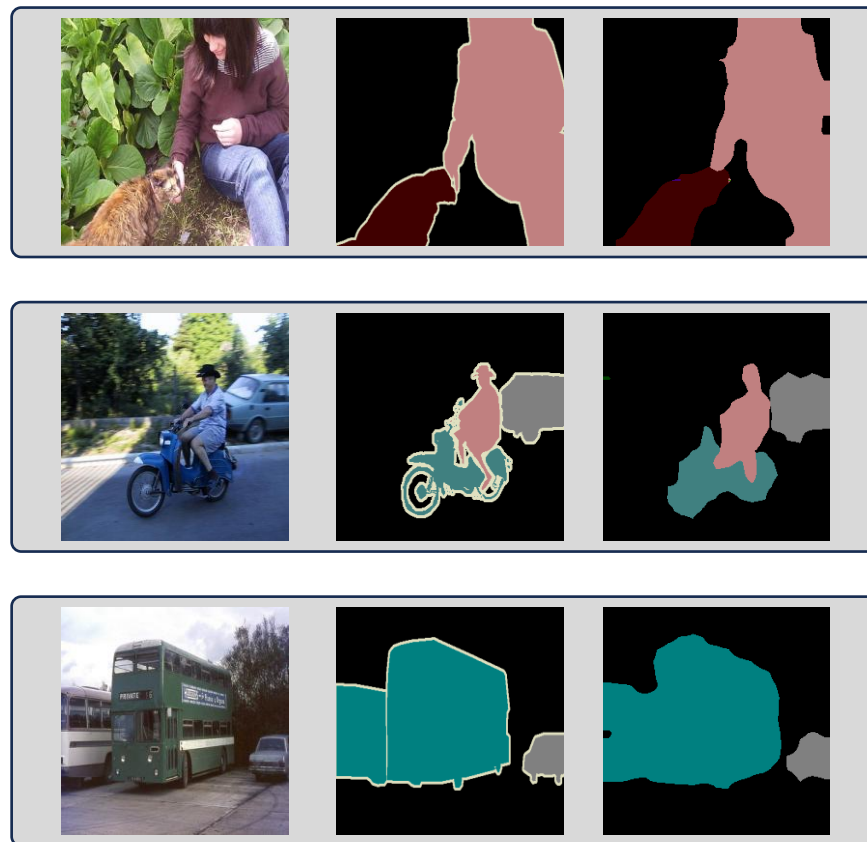
▪ Classification & semantic segmentation

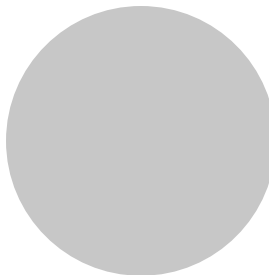
	Classification
Accuracy	84.16%

Classification



	Segmentation
mIOU	78.87%





The end

Thanks

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